Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Scalable Distributed Systems · Cloud (AWS/Azure) · AI/ML-Enabled Products · Backend Architecture · Observability · Telemetry · Python · Microservices

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PROFESSIONAL SUMMARY

Senior Software Engineer with 8 years owning scalable, reliable, and secure distributed backend systems in production - and a track record of driving end-to-end delivery of data-intensive applications with AI/ML-enabled capabilities. At Entrata, led architecture and technical design across 20+ full-stack production modules, owned performance-critical data pipelines at 99.5% uptime across 100K+ monthly active users, and drove a Docker microservices redesign (50% scalability improvement, 30% deployment reduction). Built AgentProbe - an AI/ML-enabled observability and telemetry platform for production AI systems - and Leasel, an agentic AI backend with multi-layer reliability controls. Mentoring 5+ engineers; 300+ code reviews; M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

TECHNICAL SKILLS

Engineering & Backend: Python, SQL, FastAPI, PHP, JavaScript, TypeScript, Bash, Docker, Microservices, Scalable Distributed Systems, AWS (S3, EC2, RDS, Glue), Azure (OpenAI), Redis, RabbitMQ, REST APIs, CI/CD (GitHub Actions), Telemetry, Observability, Reliability Engineering, PostgreSQL, MySQL, Oracle, Linux, Git

AI & Agents: AI/ML-Enabled Products, LangGraph, LangChain, Agentic AI, RAG Pipelines, LLM Orchestration, ChromaDB, Model Evaluation, Hallucination Detection, Agent Observability, HITL, Prompt Engineering, OpenAI API, Anthropic API, MCP, FastAPI

ML & Data: Data-Intensive Applications, Machine Learning, PyTorch, scikit-learn, NLP, BERT, PySpark, Feature Engineering, Statistical Modeling, Tableau, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution
May 2025 - Aug 2025

AI/ML Engineer Intern

- Built a scalable agentic backend (LangGraph stateful graph) for mission planning - end-to-end architecture from NLP ingestion through conditional routing and export; reliable validation guardrails blocking non-compliant outputs before delivery.
- Implemented RAG pipeline with telemetry on retrieval quality and response groundedness; performance-optimized for latency across distributed data sources.

George Mason University
Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built data-intensive ML pipelines (Python, R) on WHO and UK Road Safety datasets - $R^2 = 0.977$; full pipeline from data ingestion through feature engineering, model optimization, and statistical validation across 190+ countries.
- Mentoring 600+ students in Python and database design; designed Tableau dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS
Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Led architecture and technical design of 20+ full-stack production modules** - drove end-to-end delivery across 120+ Agile sprints with 99.5% uptime SLA at 100K+ monthly active users; zero compliance violations across 7 years of ownership of data-intensive backend systems.
- **Designed and evolved scalable distributed backend architecture** - Docker microservices redesign (50% scalability improvement, 30% deployment reduction); RabbitMQ event-driven processing with retry logic and idempotency guarantees; Redis caching for high-performance data access.
- **Built telemetry and observability tooling** - internal debugging infrastructure reducing triage time 40%; performance monitoring across multi-tenant production environments; reliability engineering maintaining sub-10 bug count for 6+ consecutive months.
- **Cloud infrastructure and security** - AWS S3, EC2, RDS at production scale; 100% PHI compliance with Libsodium-encrypted pipelines (zero security incidents); scalable data processing pipelines (fragmented federal data → 180,000 domain records per release cycle).
- Mentoring 5+ junior engineers; 300+ peer code reviews; 200+ SQL query optimizations (30% performance gain) through execution plan analysis, indexing, and stored procedure refactoring.

Entrata India - Pune
Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built property management backend modules (PHP, Python, PostgreSQL); data-intensive pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: AI/ML-Enabled Observability & Telemetry Platform

Observability · Reliability

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) | github.com/venkatalakshmiparimala/AgentProbe

Scalable AI/ML-enabled backend: 15-endpoint FastAPI service, 5-metric telemetry engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated reliability gate blocking regressions before merge, LLM-as-judge quality scoring, and multi-provider routing for performance optimization. Data-intensive evaluation across 1,240+ test cases with full observability into model behavior.

Leasel: Agentic AI Backend Platform (Distributed, Reliable)

AI/ML · Backend

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Scalable distributed AI backend: 8-agent stateful graph (LangGraph) with conditional routing, multi-source data ingestion, and reliability controls - HITL escalation when model confidence is too low. 93.6% precision, 87.9% F1 across 1,240 data-intensive test cases; CI reliability gate with telemetry-driven quality scoring.

AI on FHIR: NLP-Driven Medical Records Query System

NLP · Backend

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Scalable NLP backend translating natural-language queries into FHIR API calls using LLMs and BERT embeddings; reduced retrieval time ~60%; secure RBAC architecture across structured medical records with full data isolation.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017